

Deep Learning with Multi Modal Ensemble Fusion for Epilepsy Diagnosis

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Abstract- Epilepsy is a neurological condition affecting the brain and nervous system. Major complications and sometimes death also result, if the problem is not diagnosed and treated earlier. Most of existing works on automated Epilepsy detection are based on EEG features alone. Different from these works, this work proposes a multi modal fusion approach with deep learning-based features on each data stream and an ensemble fusion on those results. With multi modal ensemble fusion, the accuracy of Epilepsy diagnosis increases.

Index Terms— Deep Learning, Multi modal Ensemble Fusion, Epilepsy diagnosis

I. INTRODUCTION

In layman terms Epilepsy is simply called as Fits. Epilepsy is neurological illness affecting the brain. Neurons in brain use electrical signals for communication. These electrical signal communication forms the basis for various neural activities of living beings. Failure in communication at some point, results in seizure of brain activities. This phenomenon is termed as epileptic seizure or fits. Some of the patients may lose awareness during this time. They may not respond to the bystanders command. After the seizure is over they regain their consciousness level. The results of this seizure are stopping of speech, motor seizure etc. Patient himself /herself is not aware of this pause time. Such a seizure is called as minor epileptic seizure. If this attack becomes worse, it results in losing of consciousness and the whole body starts jerking vigorously. This condition is called major epileptic seizure. Aura is a condition where patients gets feeling of upcoming seizure. This disorder affects a very large number of people worldwide. Due to brain abnormalities, seizures or periods of unusual behavior results. It also results in sensations, and sometimes loss of awareness. It is characterized by recurrent unprovoked seizures. Timely diagnosis and treatment of Epilepsy is important to prevent fatal death and other severe problems due to Epilepsy. Based on various factors like type of seizure, onset age and possible etiology, Epilepsy is categorized to following types.

Localization-related epilepsies	Seizures caused due to it have a focal or partial onset.
Idiopathic epilepsies	It occurs without any identifiable pathologic cause
Symptomatic epilepsies	Caused with an identifiable symptom.
Epilepsy syndromes	It generally starts at earlier age and develops through adulthood.

Computer aided diagnosis has become the important approach towards timely detection of Epilepsy. The current approaches for Computer aided Epilepsy detection is in three categories

1. EEG based diagnosis
2. Video Event based diagnosis
3. ECG correlation based diagnosis

The goal of any computer aided diagnosis method is twofold- higher accuracy and lower false positives. The accuracy can be improved by using the information from multiple streams instead of relying on features from individual streams alone. This proposed solution in this work is based on multi modal ensemble fusion of classifier results built on each individual information streams. Due to multi modal ensemble fusion, the accuracy of diagnosis increases. Following are the important contributions in this work

1. Deep Learning features extracted and used for classification instead of user defined features.
2. Multi modal weighted ensemble fusion of individual classifier results reduces the false positives
3. Use of continuous data instead of snapshot data, thereby trend of data is used for classification. Building classifier based on trend of data increases the accuracy.

II. RELATED SURVEY

In [1], seizure detection using EEG was proposed. Dual tree complex wavelet transformation is executed on EEG signal

TABLE I. EPILEPSY TYPES

to extract features. Energy and entropy-based features are calculated from the wavelet transformed coefficients. These features are used to train the general regression neural network. The method is able to achieve ceiling level classification performance on ictal and non-ictal datasets. Due to manual selection of features, the accuracy is not high for all real datasets. Authors in [2] proposed an epileptic seizure detection framework. The framework used multiple time-frequency analysis for feature extraction and integrated random forest model with grid search optimization is used for classification. The frequency analysis is carried out using Short time Fourier transformation. Feature dimension is reduced with Principal component analysis. Random forest classifier is trained using optimal features. The optimal sets of training parameters for the random forest are found using grid search optimization to improve the accuracy. The accuracy could not be increased to greater value due to use of snapshot data instead of continuous trend for feature extraction. In [3] author proposed two approaches for EEG seizure detection based using (SpPCA) subpattern based PCA and (SubXPCA) cross-subpattern correlation-based PCA. Feature extraction can be done by using these PCA algorithms. (SVM) Support Vector Machine is trained with the extracted features using PCA algorithms. The correlation between the sub patterns in the EEG signal guides the decision process. Radial bias kernel SVM was used for classification. Early stages of seizures cannot be detected in this approach. Similar to [3] Author in [4] proposed a seizure detection approach. The feature extraction can be done by the approach tunable-Q wavelet transform (TQWT). The extracted features are used to train the SVM for classification. Features extraction is done using Authors in [5] used discrete wavelet transform (DWT) for feature extraction. Sub band wavelet coefficients are obtained applying DWT. Features are extracted from the sub band coefficients by calculating the singular value in the covariance matrix. (M-LEM) Multiplicative extreme learning machine is trained based on features and used for classification process. M-ELM has higher accuracy than SVM. A novel method for EEG seizure detection is proposed in [6]. MRBF (Multiscale Radial Basis Function) - MPSO (Modified Particle Swarm Optimization) framework is used to extract time frequency features. PCA is used for reducing the feature dimension and select optimal features. SVM classifier with Radial Bias Function kernel is trained to separate EEG seizures. Work in [7] used a new feature called Kraskov entropy extracted from EEG signal. The feature is extracted by applying tunable-Q wavelet transform on the EEG signal. The features are used to train a least square support vector machine (LS-SVM) for classifying EEG seizures. In [8] authors proposed a new feature extraction technique called wavelet-based directed transfer function (WDTF). This method integrated wavelet decomposition and the directed transfer function (DTF)

algorithms. The salient part in this method is that it can be applied to long term EEG recordings. Different from previous approaches on snapshot or short duration EEG recording, this approach used long terms EEG recordings for feature extraction. In [9] authors used deep CNN model for detection of EEG seizures. Spectral, temporal features are extracted with deep learning model and model is the model is used to learn general structure of seizure. Instead of hand-crafted features, authors in [10] used Convolutional neural network (CNN) to learn signal characteristics for epileptic seizures detection. In [11] authors represented the EEG data in spatial temporal form and this data is input to a CNN modal for EEG pathology detection. For classification different types of seizure authors in [12] implemented a deep Convolutional neural network (having 13 layers). It classifies normal, preictal, and seizure EEG signals. In [13] author proposed integrated signal analysis for seizure detection. The author used (PPG) photoplethysmography and (ECG) electrocardiography and with EEG. The method was tested for detecting seizures in temporal lobe epilepsy (TLE) patients. Feature extracted from heart rate and heart rate increase are used as indicator for epilepsy seizures. This work reduced the false positive ratio in detection of epilepsy seizures by augmenting EEG with ECG and PPG observations. Similar to [13] authors in [14] extracted heart rate characteristics from ECG signals and trained a SVM classifier to classify epileptic seizures. Augmenting ECG features based detection with EEG based detection is found to reduce the false positives.

Deep learning was used to automatically learn the discriminative EEG features of epileptic seizures in [15]. The EEG time series is first segmented to sequence of non overlapping epochs. A high level representation of features is learnt using (LSTM) network Long Short-Term Memory. The features then passed to Softmax function for training and classification. Deep learning network using EEG signals for seizure detection is proposed in [16]. It used Convolutional neural network to extract features. The extracted features are used to train a bidirectional recurrent neural network to predict the seizure. The advantage in this method is that it can predict seizures earlier than the current methods. Both EEG and ECG signals are used for early detection of seizures in [17]. Various signal processing methods such as spectral analysis, statistical methods and Empirical Mode Decomposition are used for extracting various features out from ECG and EEG frequency signals to learn rules for detection of seizure. Authors in [18] proposed an efficient multi-domain fusion method. The method combined various algorithms like Yule Walker to Autoregressive method to extract features from EEG signal. An ensemble classifier is trained with the extracted features. The classifiers used for ensemble are (LDA) Linear Discriminant Analysis classifier, (SVM) Support vector machine classifier, (KNN) K-nearest Neighbor, (ANN) Artificial Neural Network classifier. Applying fusion methods for extraction of features and classification of features is found to have increased accuracy on detection

of epileptic seizures. The table 2 summarized literature survey.

TABLE II. SURVEY SUMMARY

Paper Link	Brief	Demerit
[1]	Feature Extraction: Dual tree complex wavelet transform Classifier: Neural network	Due to manual selection of features, the accuracy is not high for all real datasets
[2]	Feature extraction: Multiple time-frequency analysis Classifier: Integrated random forest model along grid search optimization.	Accuracy could not be increased to greater value due to use of snapshot data instead of continuous trend for feature extraction
[3]	Feature extraction: PCA Classifier: Support Vector Machine (SVM)	Early stages of seizures cannot be detected in this approach
[4]	Feature extraction: Tunable-Q wavelet transform (TQWT) Classifier: SVM	Without feature selection, the accuracy is low
[5]	Feature extraction: DWT Classifier: extreme learning machine	The accuracy of ELM is limited due to manual selection of features
[6]	Feature extraction: MRBF + PCA Classifier: SVM	Early stages of seizures cannot be detected in this approach
[7]	Feature extraction: Kraskov entropy derived from Tunable-Q wavelet transform Classifier: SVM	Without feature selection, the accuracy is low
[8]	Feature extraction: wavelet decomposition Classifier: Statistics	Limited capability for continuous data analysis
[9]	Classifier: Deep CNN	Spectral, temporal features are extracted only at higher level
[10]	Classifier: CNN	Feature extraction method is limited
[11]	Classifier: CNN with Spatial closeness	Lacks support for continuous data analysis
[12]	Classifier: CNN with 13 layers.	Training and classification time is higher
[13]	electrocardiography (ECG) and photoplethysmography (PPG) is augmented with EEG	Feature selection is manual
[14]	heart rate characteristics from ECG signals and trained a SVM classifier to classify epileptic seizures	The augmentation procedure for ECG with EEG is not generic.
[15]	Classifier: Long Short-Term Memory (LSTM) network	Noise artifacts affects accuracy
[16]	Classifier: CNN with Spatial features.	Lacks support for continuous analysis

III. INFERENCES FROM THE LITERATURE SURVEY

From the study of related works following observations are made.

1. Most methods are based on user selected features
2. Most of the methods are based only on EEG analysis and very few methods use ECG augmentation to reduce false positives.
3. Most of the methods are based on Snapshot data, due to which the time to predict seizure is very short.
4. Existing multi fusion methods are based only on ensemble at classification stage.

The proposed solution can be build based on following observations.

1. Use of deep learning methods to learn the features help to identify the best features correlating to detection of seizures.
2. Fusion of other data streams like ECG with EEG based Classification can reduce the false positives.
3. Classifier built on trends associated with long term data can help forecasting the seizure earlier.
4. Fusion of multiple methods during feature extraction can increase the accuracy of detection.

Performance Measures

The performance of the proposed multi modal ensemble fusion is compared with [17] and [15]. Work in [17] used both ECG and EEG for epileptic seizure prediction and work in [15] applied deep learning feature extraction on EEG for epileptic seizure prediction. The performance of these three solutions is measured and compared in terms of following:

TABLE III. TYPES OF CLASSIFICATION ERRORS

True Label	Predicted Label	Type of Error
True	True	True Positive (TP)
False	False	True Negative (TN)
False	True	False Positive (FP)
True	False	False Negative (FN)

1. Accuracy:

$$Accuracy = \frac{\# \text{ correct predictions}}{\# \text{ total preiccions}}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

2. Sensitivity:

$$Sensitivity = \frac{TP}{TP + FN}$$

3. Specificity

$$Specificity = \frac{TN}{TN + FP}$$

4. False Positive Rate:

$$FPR = \frac{FP}{TN + FP} = 1 - Specificity$$

Five-fold cross validation with ratio of 70:30 for training and testing is used to measure performance.

IV. CONCLUSION

Increasingly large numbers of people all over the world are getting affected by neurological disorder called Epilepsy. Electrical activity of brain is captured using sensors and Deep learning is applied to analyze it and predict the seizure ahead of its occurrence. A multi modal ensemble fusion method is proposed for early prediction of seizures. The work used multiple classifier models using EEG and ECG signals to build four different

classifiers. Two of the classifier used deep learning for both feature extraction and classification. Other two classifiers used deep learning only for classification. The proposed solution will have higher accuracy and reduced false positives. Due to its reliance on continuous data, the approach is able to predict seizure much earlier.

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